Health & Wellness AI Agent - Project Report

Executive Summary

The **Health & Wellness AI Agent** is an advanced, holistic, empathetic conversational assistant designed to help users achieve both their physical and mental well-being goals. The application parses natural language inputs and uses a robust agentic architecture to analyze goals, suggest meal and workout plans, recommend stress-relief techniques, schedule check-ins, and track progress.

The system leverages OpenAI-compatible language models (via OpenRouter), a FastAPI backend, and an intelligent tool/sub-agent routing system to provide highly personalized health guidance.

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1. System Architecture

1.1 Backend Framework

- **FastAPI**: Serves as the high-performance backend web framework (`app.py`). It exposes RESTful API endpoints (like `/api/chat` and `/api/reset`) to communicate with the frontend.
- **LLM Integration**: Uses an `AsyncOpenAI` client initialized through OpenRouter (`openrouter_api_key`) to communicate with state-of-the-art models.
- **Session Management**: Manages user conversations in memory, tracking chat history and contextual state (goals, meal plans, workout plans) to maintain context over multiple conversational turns.

1.2 Context & Guardrails

- **User Session Context (`context.py`)**: A centralized state object that holds the user's ongoing data, including:
- `goal`
- `meal_plan`
- `workout_plan`
- `relaxation_plan`
- `progress_logs`
- `checkin_schedule`
- **Output Validation (`guaidrails.py`)**: Employs **Pydantic** models to rigorously validate inputs and enforce strict JSON schemas for the agent's outputs, ensuring tools always receive and return predictably structured data (e.g., `MealPlan`, `WorkoutPlan`, `Goal`).

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2. Agentic Workflow & Routing

The system uses a multi-agent orchestration pattern where a primary agent handles general requests and routes complex scenarios to specialized sub-agents or functional tools.

2.1 Main Agent (`agent.py`)

- **Name**: `HealthWellnessPlanner`
- **Role**: Empathetic, conversational planner focusing on comprehensive health (physical + mental).
- **Core Loop**:
- Parses natural, conversational requests.
- Identifies specific needs (Diet, Workout, Stress Management).
- Orchestrates internal tools to build the user's personalized wellness plan.

2.2 Specialized Handoff Agents (`my_agents/`)

When the Main Agent detects a situation beyond its primary scope, it triggers a handoff to a specialized agent: 1. **Nutrition Expert Agent (`nutrition_expert_agent.py`)**: Invoked for complex dietary needs, food allergies, or medical dietary restrictions like diabetes. 2. **Injury Support Agent (`injury_support_agent.py`)**: Invoked when the user mentions physical limitations, chronic pain, or injuries (e.g., knee pain). 3. **Escalation Agent (`escalation_agent.py`)**: Triggered when the user requests direct communication with a human health coach or professional.

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3. Tool Ecosystem (`tools/`)

The Main Agent has access to a robust suite of Python-based tools to execute specific tasks:

1. **Goal Analyzer (`goal_analyzer.py`)**
2. Parses unstructured human language (e.g., "I want to lose 5kg in a month") into structured goal tracking metrics.
3. **Meal Planner (`meal_planner.py`)**
4. Generates 7-day meal plans based on dietary preferences.
5. **Internet Search Integration**: Actively searches the internet (via DuckDuckGo) for real, up-to-date healthy recipes.
6. **Smart Fallback**: If the internet search times out or fails, it intelligently falls back to a massive predefined library of real-world meals (e.g., "Cucumber Salad", "Avocado Turkey Wrap") instead of using generic placeholder names.
7. **Workout Recommender (`workout_recommender.py`)**
8. Crafts targeted exercise routines based on the user's fitness goals.
9. **Relaxation Recommender (`relaxation_recommender.py`)**
10. Addresses the mental/emotional side of health. Suggests stress-free living strategies, mindfulness routines, and sleep hygiene improvements based on detected stress or anxiety.
11. **Scheduler (`scheduler.py`)**
12. Sets up recurring check-ins and accountability schedules for the user.
13. **Progress Tracker (`tracker.py`)**
14. Logs metrics over time to help the agent track whether the user is successfully meeting their baseline goals.

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4. Frontend & Deployment

- **Web UI (`static/index.html`)**: The application serves a dedicated chat interface from the `/static` directory where users interact with the agent in real-time. The UI displays both the conversation history and a live-updating "Context Panel" showing their active goals and generated plans.
- **Vercel Compatibility**: The codebase is optimized for serverless deployments (specifically Vercel), with logic in `app.py` specifically tailored to route static files correctly within Vercel's Lambda and Edge bundle structures.

Conclusion

The Health & Wellness AI Agent represents a complete, modern AI application. By combining strict output guardrails, an advanced toolset (with real-time internet search), specialized sub-agents, and a focus on both physical and mental well-being, the agent successfully delivers personalized, realistic, and actionable health guidance.